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import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split, KFold, cross_val_score, GridSearchCV
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
from sklearn.preprocessing import StandardScaler
print("--- Part 1: Hyperparameter Tuning with Grid Search ---")
iris = load_iris()
X = iris.data
y = iris.target
feature_names = iris.feature_names
target_names = iris.target_names
print("\nDataset Features (X) shape:", X.shape)
print("Dataset Labels (y) shape:", y.shape)
print("Feature Names:", feature_names)
print("Target Names:", target_names)
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.3,
    random_state=42,
    stratify=y
)
print("\nTraining set size:", X_train.shape[0], "samples")
print("Test set size:", X_test.shape[0], "samples")
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
print("\nFeatures standardized.")
param_grid = {
    "C": [0.1, 1, 10, 100],
    "gamma": [1, 0.1, 0.01, 0.001],
    "kernel": ["rbf", "linear"]
}
print("\nHyperparameter grid defined:")

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    param_grid=param_grid,
    cv=5,
    scoring="accuracy",
    verbose=1,
    n_jobs=1
)
print("\nStarting Grid Search with 5-fold Cross-Validation...")
grid_search.fit(X_train_scaled, y_train)
print("\nGrid Search completed.")
print("\nBest hyperparameters found:")
print(grid_search.best_params_)
print(
    "Best cross-validation accuracy:",
    round(grid_search.best_score_, 4)
)
best_model = grid_search.best_estimator_
y_pred_tuned = best_model.predict(X_test_scaled)
test_accuracy_tuned = accuracy_score(
    y_test,
    y_pred_tuned
)
print(
    "\nTest set accuracy with tuned model:",
    round(test_accuracy_tuned, 4)
)
print("\n--- Classification Report for Tuned Model ---")
print(
    classification_report(
        y_test,
        y_pred_tuned,
        target_names=target_names
    )
)
print("\n--- Confusion Matrix for Tuned Model ---")
cm_tuned = confusion_matrix(
    y_test,
    y_pred_tuned
)

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)
plt.title("Confusion Matrix (Tuned SVM)")
plt.xlabel("Predicted Label")
plt.ylabel("True Label")
plt.show()
results_df = pd.DataFrame(
    grid_search.cv_results_
)
print("\n--- Top 5 Grid Search Results ---")
top_results = results_df[
    [
        "param_C",
        "param_gamma",
        "param_kernel",
        "mean_test_score",
        "rank_test_score"
    ]
].sort_values(
    by="rank_test_score"
).head()
print(top_results)
print("\n--- Part 2: Cross-Validation Techniques (K-Fold) ---")
model_cv = SVC(random_state=42)
k_folds = 5
kf = KFold(
    n_splits=k_folds,
    shuffle=True,
    random_state=42
)
print(
    "\nPerforming",
    k_folds,
    "-fold cross-validation..."
)

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print(
    "\nCross-validation scores for each fold:",
    cv_scores
)
mean_cv_accuracy = np.mean(cv_scores)
std_cv_accuracy = np.std(cv_scores)
print(
    "Mean cross-validation accuracy:",
    round(mean_cv_accuracy, 4)
)
print(
    "Standard deviation of cross-validation accuracy:",
    round(std_cv_accuracy, 4)
)
plt.figure(figsize=(8, 5))
plt.bar(
    range(1, k_folds + 1),
    cv_scores
)
plt.axhline(
    y=mean_cv_accuracy,
    linestyle="--",
    label=f"Mean Accuracy ({mean_cv_accuracy:.4f})"
)
plt.title(
    f"{k_folds}-Fold Cross-Validation Accuracy Scores"
)
plt.xlabel("Fold Number")
plt.ylabel("Accuracy")
plt.ylim(0.8, 1.0)
plt.legend()
plt.grid(
    axis="y",
    linestyle="--"
)

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print(
    "Best cross-validation accuracy:",
    round(grid_search.best_score_, 4)
)
best_model = grid_search.best_estimator_
y_pred_tuned = best_model.predict(
    X_test_scaled
)
test_accuracy_tuned = accuracy_score(
    y_test,
    y_pred_tuned
)
print(
    "\nTest set accuracy with tuned model:",
    round(test_accuracy_tuned, 4)
)
print("\n--- Program Completed Successfully ---")

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*** --- Top 5 Grid Search Results ---

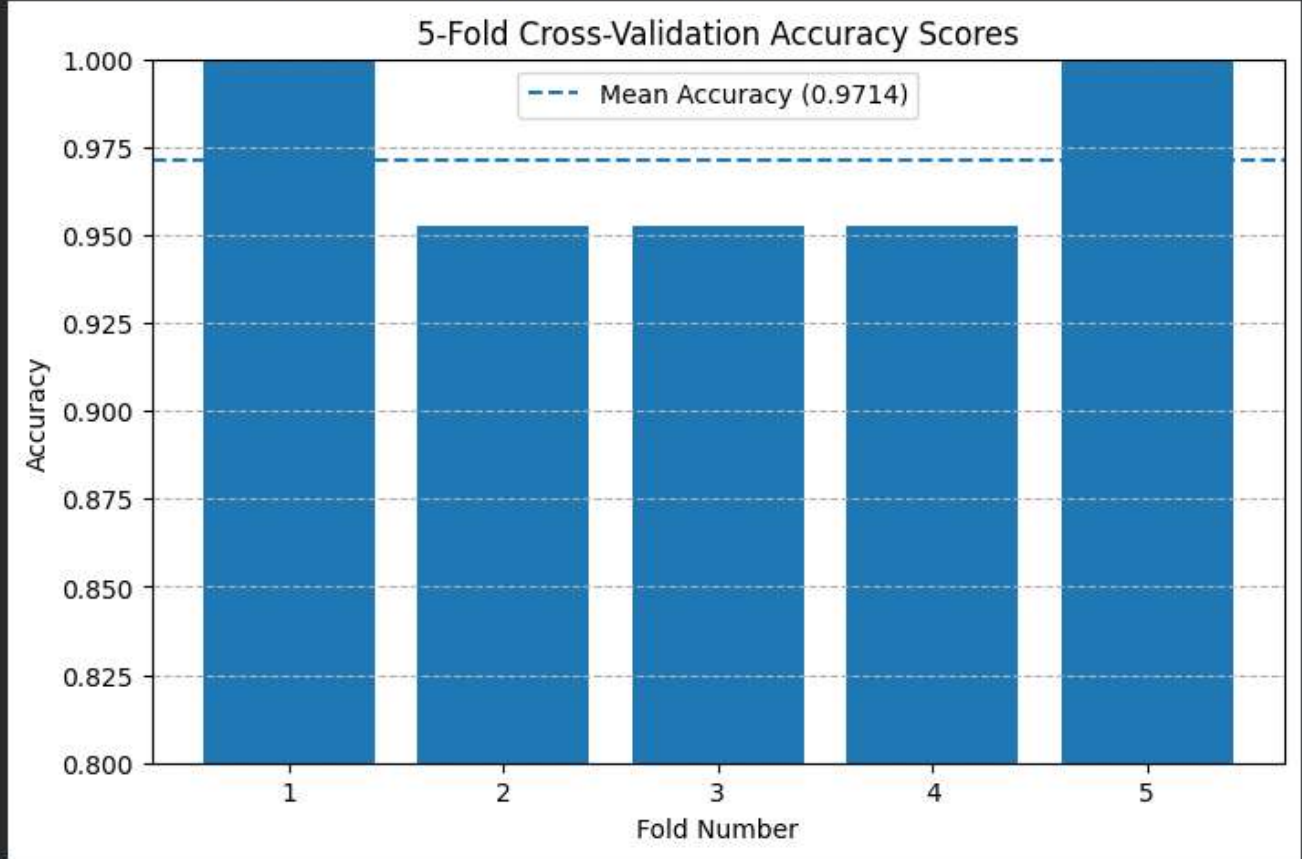
	param_C	param_gamma	param_kernel	mean_test_score	rank_test_score
10	1.0	0.100	rbf	0.980952	1
27	100.0	0.100	linear	0.980952	1
31	100.0	0.001	linear	0.980952	1
29	100.0	0.010	linear	0.980952	1
25	100.0	1.000	linear	0.980952	1

--- Part 2: Cross-Validation Techniques (K-Fold) ---

Performing 5 -fold cross-validation...

Cross-validation scores for each fold: [1. 0.95238095 0.95238095 0.95238095 1. 1]

Standard deviation of cross-validation accuracy: 0.0233



--- Final Grid Search Results ---

Starting Grid Search with 5-fold Cross-Validation...
Fitting 5 folds for each of 32 candidates, totalling 160 fits

Grid Search completed.

Best hyperparameters found: {'C': 1, 'gamma': 0.1, 'kernel': 'rbf'}